

BIOS232 – Assign I (Part 1 of 2): Short Critical Review

Muth, F. (2021). Intra-specific differences in cognition: bumblebee queens learn better than workers. *Biology letters*, 17(8), 20210280.

This work will serve as a critique of Dr Felicity Muth's 2021 paper "Intra-specific differences in cognition: bumblebee queens learn better than workers". Bumblebees are common model systems in cognitive ecology. Despite this, there did not seem to be an intra-specific (caste-based) characterisation of learning abilities, and it is this important research gap that Muth's article fills. The study establishes a difference in cognitive performance between queens and female foragers and ascribes it to ecological roles. However, given the methodological reliance on multiple *Bombus* species, a greater acknowledgment of interspecific effects would strengthen both the study's discussion and experimental design.

Muth motivates the study by noting that honeybees and bumblebees are model systems for learning, and that cognitive abilities can be shaped by natural selection. The key gap is that, despite *Bombus* being widely studied, most work focuses on a single caste (i.e., female foragers), with little to no direct work on queens. Additionally, much of the learning literature relies on lab-reared animals rather than wild learning. The central hypothesis is that queens should outperform workers because mistakes are "costlier" for solitary colony-founding queens as compared to foragers because the future reproductive success depends on efficient early foraging by a single individual.

To test floral association, wild *Bombus vosnesenskii* queens and workers were collected across desert and meadow sites and trained to associate a rewarding colour with sucrose. Bees

received initial no-choice exposures to the rewarding and unrewarding stimuli and then completed choice trials where a “correct” response was choosing the previously rewarded colour first; a subsequent unrewarded test phase measured preference when both options contained water. Learning (correct/incorrect choices) was analysed using binomial GLMMs, and motivation was proxied via sucrose responsiveness (included in analysis).

Across both desert and meadow sites, queens showed a higher proportion of correct choices than workers in learning and test phases. Bees trained to yellow outperformed blue, with a stronger colour preference effect in the meadow. To extend the evolutionary argument, brood-parasitic *Bombus insularis* queens were tested and performed similar to non-parasitic queens. To control for foraging experience, a lab assay compared *Bombus impatiens* unmated queens (gynes) versus workers; because bees learned too readily under the field protocol, the discrimination was made harder (yellow replaced with lighter blue/aqua and fewer choice trials). The paper suggests that lab-reared bees may be more “ingenuous” to colour cues and may “overcommit” to signals due to limited prior experience. Here too, the gynes outperformed workers. Muth also notes that queen’s sensory advantages could be the reason for better performance, as well as suggests a possible evolutionary advantage of ‘reduced’ learning abilities in foragers. Muth also situates the findings within broader questions about caste, life stage, and ecological specialization in *Bombus* cognition.

From a critical perspective, the introduction explicitly highlights the motivation for the study and the research gap it addresses. The hypothesis, i.e., queens outperform workers because mistakes are ‘costlier’ for solitary colony-founding queens, was precisely scoped, falsifiable, and reasonably derived from tropes in cognitive ecology. Though the hypothesis itself was not too conceptually ‘shocking’, it is novel in terms of the system to which these tropes are applied.

Even if the prediction appears intuitive, systematically testing such systems remains worthwhile, as rigorous empirical work may contradict expectations and thereby advance the field. The methodology developed to test floral association abilities was also novel and methodologically careful, using justifiable proxies for learning (proportion of ‘correct’ choices) and motivation (‘sucrose responsiveness’ – which was controlled for in the statistical tests).

Furthermore, the replication of results derived from the wild population in the lab-reared species represents a strong aspect of the study. In addition to corroborating the hypothesis, this observation helps constrain potential confounders. The reproducibility across distinct growth environments lends support to the use of lab-reared bees for future cognition research. Though the author briefly acknowledges that the performance gap between lab-reared and wild bees may reflect species-level variation. However, without citing evidence it is claimed that it is more likely that the gap be ascribed to the difference in floral-association noise experienced. A related interspecific critique applies to the use of the parasitic queen example. Muth employs this comparison to strengthen the evolutionary logic underlying why queens exhibit superior associative learning, and the approach is both creative and conceptually elegant. At the same time, it does not fully account for interspecific cognitive variation. Both these instances are inconsistent with some minimal evidence indicating that *Bombus* species differ in cognitive performance (Muth et al., 2018). The parasitic-queen comparison would therefore require benchmarking against actively foraging castes of the same species to satisfactorily corroborate Muth’s hypothesis. The discussion section adopts an appropriately cautious tone, which is valuable because it avoids reducing the findings solely to learning and acknowledges perceptual advantages as a plausible alternative explanation.

Muth's study investigates caste-based cognitive differences in bumblebees, addressing a gap in *Bombus* research by examining queens alongside female foragers. The work presents a careful experimental demonstration of performance differences grounded in ecological reasoning, supported by structured learning assays and cross-context replication. While the findings are methodologically strong and the discussion remains appropriately cautious, greater consideration of interspecific effects would strengthen the interpretive framework. Overall, the article is a valuable and well-executed contribution that advances questions about caste, life stage, and ecological specialization in cognitive ecology.

Other References

Muth, F., Cooper, T. R., Bonilla, R. F., & Leonard, A. S. (2018). A novel protocol for studying bee cognition in the wild. *Methods in Ecology and Evolution*, 9(1), 78-87.